

L11: Coevolutionary Effects in Twitter Retweets

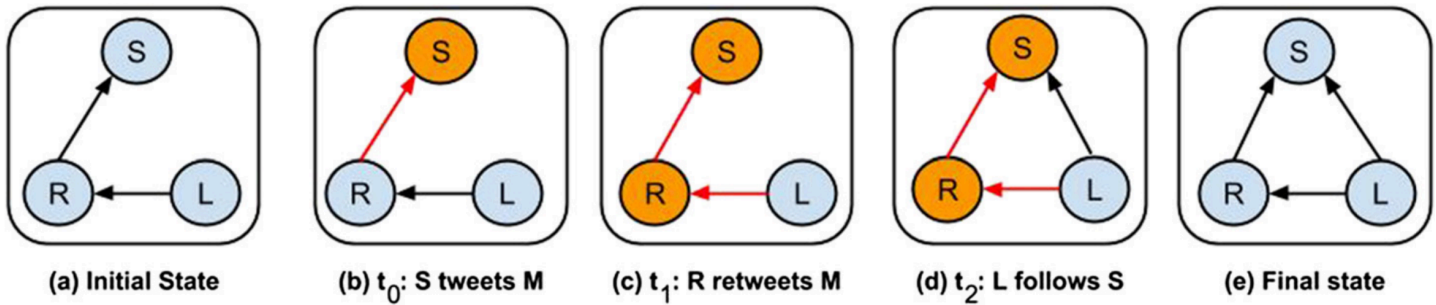


Image Source: [“Co-evolutionary dynamics in social networks: a case study of Twitter”](https://link.springer.com/article/10.1186/s40649-015-0023-6) [by Antoniadis and Dovrolis](https://link.springer.com/article/10.1186/s40649-015-0023-6)

Coevolutionary effects have also been observed empirically, on social networks.

Think about Twitter. Consider three users: S (*speaker*), R (*repeater*) and L (*listener*) – see the visualization. Initially, R follows S and L follows R. Suppose that R retweets a post of S. The user L receives that message and it may decide to follow user S. When L starts following S, the network structure changes as a result of an earlier information transfer on the network.

Additionally, this structural change will influence future information transfers because L will be directly receiving the posts of S.

How often does this happen? The study by Antoniadis and Dovrolis referenced here analyzed a large volume of Twitter data, searching for the addition of new links from L to S shortly after the retweeting of messages of S from R.

In absolute terms, the probability of such events is low: between 10^{-4} to 10^{-3} , meaning that only about 1 out of 1000 to 10000 retweets of S lead to the creation of new links from a listener L to S.

Even though these adaptation events are infrequent, they are responsible for about 20% of the new edges on Twitter.

Additional statistical analysis showed that the probability of such network adaptations increases significantly (*by a factor of 27*) when the reciprocal link, from S to L, already existed.

Another factor that increases the probability (*by a factor of 2*) that L will follow S is the number of times that L receives messages of S through any repeater R.

Such network adaptation effects have profound effects on information diffusion on social networks such as Twitter because they create **“echo chambers”** – a positive feedback loop in which a group of initially loosely connected users retweet each other, creating additional links within the group, and thus, making future retweets and new internal links even more likely.